AI-FIRST SDLC PLATFORM

Platform Overview, Phase-by-Phase Build, & Local Execution Guide

An engine-agnostic, multi-tenant, web-delivered agentic platform that runs the full software lifecycle — ideation through deployment — on a client's real tools of record, with a human review gate at every state change.

Prepared for	The Engineering Team — EdvenSoft
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1. What We Are Building — and What We Are Trying to Achieve

The product is not “an AI that writes code.” It is a platform that runs your entire software lifecycle the way a disciplined engineering organisation does — with specialists, evidence, review gates, and an audit trail — but with AI agents doing the heavy lifting and humans staying in control of every decision that matters.

1.1 The ultimate goal

We already had a powerful asset: a suite of **17 expert AI agents** (ideation, requirements, PRD, architecture, UX, task-breakdown, implementation, code review, testing, security, CI/CD, deployment, and more) plus the rules and gates that made them behave like senior engineers. The catch: that suite ran *locally*, glued to one developer's machine and Git checkout, with project context living in Markdown files.

The goal of this platform is to transform that suite into a **multi-tenant, web-delivered, engine-agnostic agentic SDLC platform** that:

- operates each client's **real tools of record** — Confluence, Jira, Figma, Xray, GitHub — bidirectionally, instead of inventing parallel copies;
- runs the **full lifecycle** (idea -> requirements -> PRD -> architecture -> UX -> build -> review -> test -> security -> CI/CD -> deploy) for both new (greenfield) and existing (brownfield) projects, single- or multi-repo;
- keeps a **human review gate** at every state change — nothing reaches a system of record without an approval;
- is **durable** (a run survives crashes, restarts, and an offline laptop) and **reproducible** (every run records the agent version, model, inputs, outputs, cost, and approver);
- treats the AI **engine as swappable** — Claude today, others later — without rewriting a single agent.

> **In one sentence**
>
> Take the intelligence of 17 senior-engineer AI agents, lift it off the laptop into a governed cloud-ready platform, point it at the client's real tools, and let teams ship software faster without giving up control, traceability, or quality.

1.2 The seven non-negotiables (the rules the whole system obeys)

Every design decision serves these locked principles. They are worth internalising because they explain *why* the platform is shaped the way it is.

#	Principle	What it means in plain terms
1	State vs. content	The platform owns **state** (pointers, provenance, events in Postgres). External tools own **content** (the actual Jira issue, Confluence page, Git PR). Local files are throwaway.
2	Orchestration vs. execution	The platform decides *what* happens and *when* (and runs the gates). The AI engine only *executes* a step. So engines stay interchangeable.
3	Memory = index, not copies	We store pointers + hashes + versions of artifacts, never copied bodies. This is what lets us detect drift and cross-check conflicts.

#	Principle	What it means in plain terms
4	Work is a graph	The unit of work is a **WorkItem**; work is a per-item DAG, not a rigid linear pipeline — features can progress independently.
5	Every artifact is gated	draft -> reviewed -> gated -> published. No auto-publish, ever. Nothing reaches a system of record before an approval.
6	Tenant isolation by RLS	Every database table carries a tenant_id and a Postgres Row-Level-Security policy from its first migration. One client can never see another's rows.
7	Capabilities, not vendors	Business logic calls tasktracker.* / docstore.* / codehost.* — never jira.* directly — so a connector can be swapped by editing one row.

Three more locked decisions round these out: the default engine is **Claude (Agent SDK)** behind an ExecutionDriver interface; durability is provided by **Temporal** from day one; and the agents are migrated by the **75 / 15 / 10** rule (reuse ~75% of an agent's reasoning verbatim, rewrite ~15% of its data references to the platform, replace ~10% of invocation glue).

1.3 The architecture at a glance — seven planes

The platform is organised into seven cooperating layers. You will meet each of these in the phase descriptions and again in the testing guide.

Plane	Responsibility	Where it lives
Experience	What people see and click: project workspace, phase console, gate approvals, traceability viewer.	React app (frontend/)
Control	The brain: Work-DAG state machine, Gate/Approval service, run scheduler, Manifest Composer, reconciliation, driver router.	Django services (backend/)
Execution	Runs the agents: the ExecutionDriver interface + ClaudeDriver; planning runs server-side, build runs on a developer-machine companion.	drivers/, runner/
Integration	Talks to the outside world through capability interfaces + connectors + the Platform-State MCP server + webhooks + a rate governor.	backend/integrations/, mcp_platform_state/
Memory / State	Postgres (with pgvector) + an S3-compatible object store: project profile, artifact registry, traceability graph, work DAG, event ledger, run history.	Postgres + MinIO
Security & Governance	Tenant isolation (RLS), prompt-injection defences, secret scanning, egress control, audit. (Deepens in Phase 5.)	across the stack
Observability & Quality	Durable workflow visibility, run cost metering, and the test/quality plane. (Deepens in Phase 5.)	Temporal + apps/quality/

1.4 What “done” looks like at each milestone

The build proceeds phase by phase, each ending in a working **thin vertical slice** demonstrated end-to-end. The headline milestone:

First usable product — after Phase 2

A user can take a **greenfield** project from idea -> PRD -> architecture -> backlog in Jira -> an implemented GitHub PR, and can **onboard an existing single-repo** project (from local documents or live Atlassian). Phases 3–4 add design/test/security depth and brownfield-full/multi-repo; Phases 5–6 add enterprise hardening, the cloud runner fleet, and engine choice.

2. The Build, Phase by Phase (Phases 0–4)

Each phase below lists the concrete components that were built and explains what each one does in everyday language, and closes with a **capstone slice** — the thin end-to-end demo that proves the phase works. (For a real, straight-through test pass — greenfield and brownfield — see the two journeys in **Section 5**.) Phases 0 through 4 are complete and tested.

2.1 Phase 0 — The Platform Spine

Goal: move the source of truth off the filesystem and run the existing agents through a real platform with state, gates, and provenance — before adding any external tools. This phase builds the skeleton everything else hangs on.

Component	What it is / does
State plane (Postgres)	The durable memory: Tenant, Project, **Project Profile** (the database record that replaces the old CLAUDE.md), Repository, **WorkItem** (the unit of work + its DAG), **Artifact Registry** (pointers + hashes + versions, never bodies), **Event Ledger** (append-only history), **Run History**, AgentSpec Registry.
Platform-State MCP server	A small tool server the agents call to read/write platform state: get_context, register_artifact, log_event, add_trace_edge, list/transition_work_item, etc. Every call is tenant-scoped.
Manifest Composer	Assembles a fresh, budget-bounded AGENTS.md per run from the Project Profile + relevant artifact pointers + recent events + rules. This is the context an agent reads — generated each run, never hand-maintained.
Temporal AgentRunWorkflow	The durable orchestration of one agent run: compose context -> execute agent -> verify it wrote what it should -> open the gate -> wait for a human -> publish + advance. Survives restarts; retries are idempotent.
Gate / Approval service	Gates are control-plane Approval objects, never engine hooks. A human with the right role decides; the decision signals the workflow to resume. (Phase 3 added WCAG checklist conditions to this.)
ExecutionDriver + ClaudeDriver	The seam that makes the engine swappable. Business logic calls the interface; the ClaudeDriver runs the Claude Agent SDK underneath.
Tenant isolation (RLS)	A non-superuser database role the app connects as, plus a Row-Level-Security policy on every table. Unset tenant context fails closed (sees no rows).
React Experience plane	The web UI: create a project, edit its profile, a Phase Console to trigger phases and watch the run timeline, a gate-approval screen, an artifact view, and a traceability view.
Migrated agents 01–04	Ideation, Requirements, PRD, Architecture — the first agents migrated by the 75/15/10 rule, plus the rules and per-agent triggers that replace the old slash commands.

> **Phase 0 — capstone slice (what this phase proves)**
> Run ideation -> requirements -> PRD -> architecture fully on platform state; artifacts registered as drafts in object storage; every step gated by a human; runs attributed; the manifest rendered per run. (This is exactly what you will click through in the testing guide, Section 3.)

2.2 Phase 1 — First System of Record (Atlassian)

Goal: make the platform operate a client's real Confluence and Jira, bidirectionally, with progress tracking and traceability — and let existing projects be onboarded.

Component	What it is / does
DocStore connector (Confluence)	Creates/updates Confluence pages (versioned, read-through). The platform publishes a PRD page only after its gate is approved.
TaskTracker connector (Jira)	Creates epics/stories/subtasks, transitions issues, comments, searches — headless via API token.
Capability abstraction	docstore.* and tasktracker.* interfaces + a per-tenant get_capability() selector. Callers never name Jira or Confluence directly (non-negotiable #7).
Artifact publish path	The gate -> publish -> pointer+version lifecycle to the live SoR; idempotent; falls back to “internal” when no connector is configured.
Traceability graph + viewer	requirement -> PRD -> epic -> story lineage, with deep-links into the live Confluence/Jira items.
Reconciliation service	Turns proposed work into accepted SoR issues without ever duplicating something that already exists (register-don't-recreate).
Brownfield-lite onboarding	Adopt an existing project: register what is already in Atlassian, and extract structured artifacts from static **Word/PDF/Excel** documents (with provenance + a review gate).
Rate governor	A per-(tenant, capability) token bucket + backoff + short-lived read cache, so live API calls stay within limits and don't flake.
Migrated agent 06	Task Breakdown — decomposes the PRD/architecture into a vertically-sliced backlog that reconciliation pushes into Jira on approval.

2.3 Phase 2 — Build Execution (real code, real PRs, durably)

Goal: produce real code — implementation, review, test — as GitHub pull requests, durably, *without changing how a developer works day to day*.

Component	What it is / does
CodeHost connector (GitHub)	Owns the PR / branch / status / comment / merge surface via the GitHub REST API. The engine never calls Git directly; the platform does, behind the codehost.* capability.
Local-companion runner	A small daemon a developer runs on **their own machine**. It enrolls with the platform, pulls a build task, and works in a clean, platform-managed **git worktree** (a feature/<key> branch) so the developer can run and test the change locally before any PR. Their main checkout is never touched.
Remote Platform-State serving	The companion uses the state tools over an authenticated remote transport — the dev machine needs no database credentials; tenant binding and RLS stay server-side.

Component	What it is / does
run_build_task	The build-mode driver call: launches the agent in the worktree with file + shell tools, under a per-phase tool allowlist (no CodeHost, no arbitrary network).
Temporal BuildWorkflow (saga)	Durable build orchestration: dispatch to the companion -> await its report (or park for human takeover if the laptop is offline) -> open a draft PR -> review -> remediation loop -> reviewer merge gate -> merge. Idempotent; compensating on failure.
07 <-> 08 remediation loop	Implementation (07) and Code Review (08) cooperate: each blocking finding becomes a child WorkItem; 07 re-runs in remediation mode until 08 is clean — then the merge gate can be granted.
Injection guardrails v1	Secret-scan before push (blocks credentials reaching GitHub), an egress allowlist (only approved hosts), and the per-phase tool allowlist.
Human takeover + webhook reconcile	A developer can take a WorkItem over and just work by hand; a signed GitHub webhook reconciles their commits/PR/CI status back into the registry (register-don't-recreate).
Light multi-repo	One project can target several repositories (one repo per WorkItem); each builds on its own branch/PR independently.
Migrated agents 07, 08, 11, 12	Implementation, Code Review, CI/CD, Deployment — with their reasoning (Clean Code/TDD, the eleven review dimensions, the Jenkins/Actions templates, blue-green/canary releases) preserved verbatim.

Phase 2 — capstone slice (the stop-and-dogfood MVP)

WorkItem -> companion worktree -> local build/test -> PR -> CI status -> review -> remediation -> reviewer merge gate -> merge, with a forced mid-run failure recovered idempotently, a human commit reconciled via webhook, and two repos in one project building independently.

2.4 Phase 3 — Design & Test Loops

Goal: close the UX and QA loops — read a client's live design, push tests and results to Xray, and turn a failing test into a tracked, linked bug.

Component	What it is / does
DesignSource connector (Figma)	**Read-as-SoR**: the platform reads the client's *current* Figma design at run time (frames, design tokens, components) rather than regenerating it. Canvas write-back is a deferred seam.
Design-System / tokens + Code Connect	Aggregates the current design into a design_system artifact (tokens) and a code_connect mapping that links design components to code components.
a11y (WCAG) gate	The accessibility review is enforced as a **checklist of WCAG 2.1 AA conditions on the design Approval** — an approve is blocked until every required criterion is satisfied. It is a control-plane gate, not a prompt rule.
Per-tenant theming	Design tokens + responsive breakpoints stored on the Project Profile flow verbatim into the manifest — one theme, no per-tenant code fork.
TestMgmt connector (Xray)	Authors test cases as Xray tests and imports execution results (idempotently). Headless, per-tenant, rate-governed.
Result import + defect loopback	A test run records a TestExecution + a test_report artifact; a **failing test opens a linked, traced defect** (a child WorkItem tied to the test and traced to the code); a later pass closes it.

Component	What it is / does
Test-impact selection	Given a changed piece of code, walks the traceability graph to select just the tests that cover it.
Flaky quarantine	A test that flips pass/fail within one run is quarantined and excluded from the blocking gate (never silently blocks a merge, never silently passes).
Ephemeral preview env per PR	A companion-local seam: the app is brought up from the worktree, the URL is posted on the PR, and torn down on close. (Cloud preview is a Phase-5 drop-in behind the same seam.)
Migrated agents 05, 09, 09b, 10	UX Design (read-as-SoR + HTML wireframes), Testing, Test Automation (Playwright-MCP locator discovery + Page Object Model preserved), and Security (the nine review tracks + SBOM, with criticals opening defects).

Phase 3 — capstone slice (what this phase proves)

Design produced and gated through the WCAG checklist -> implementation reads the current design -> tests author to Xray and results import -> a failing test opens a traced defect that closes on a clean re-run -> a standalone security review -> a companion-local preview env. Verified end-to-end, with the full platform suite green across all phases.

2.5 Phase 4 — Brownfield-Full & Multi-Repo

Goal: the client front door. Onboard an *existing* project — any number of repositories — understand its code, **resume the lifecycle from the middle** instead of starting over, and keep cross-repo API contracts honest with **live drift detection**. This is the entry point most real clients will use.

Component	What it is / does
CodeIntel capability + built-in analyzer	A fixed 8-tool codeintel.* surface with a **built-in SOTA analyzer** (no license needed): git/Tornhill **hotspots** (churn × complexity), a dependency graph with **PageRank** + communities, dead-code and complexity, and decision archaeology. It runs in the **companion worktree** on a real clone; the platform stores only the index + artifacts, **never the source**. Repowise (or another engine) plugs in behind the same seam.
Knowledge Index	An RLS-aware chunk store with an always-on, **zero-cost keyword (BM25) baseline** plus an optional embedding seam (Voyage / Gemini). It feeds the manifest's retrieved-knowledge section and is **project-wide shared memory** across all of a project's repos.
Agent 00c — Intelligent Context Extractor	Migrated: turns a raw codebase into five intelligence layers (dependency graph, git analytics, dead code, decisions, health) registered as platform artifacts + indexed into the Knowledge Index. tech_lead gate.
Agent 00 — Onboarding (full)	The brownfield front door: **discover** across the connected systems of record + code-intel, **register** what already exists (register-don't-recreate), **reconcile**, detect the **gaps**, and emit a **gap report** that assesses every lifecycle phase Complete/Partial/Missing and names the correct **resume phase**. product_owner gate.
Agent 00b -> API-Contract artifact	Cross-repo contracts as a **versioned api_contract registry artifact** (read-through, never a copied file): the provider repo exports it (semver-bumped); consumer repos read the current version at run time and record an implements edge. architect gate. (Agent 00d Snapshot is retired — continuity is now the database + durable runs.)

Component	What it is / does
Multi-repo project model + per-repo manifest	One project owns **N repositories** (frontend/backend/service); topology reconciles to single/split/multi. The composed manifest is **scoped per repo** (that repo's artifacts + project-shared ones + the shared contract), while the registry, traceability, and Knowledge Index stay **project-wide shared memory**.
Cross-repo DAG edges	Dependencies that span repositories: implements traceability edges across repos + WorkItem.depends_on — the basis for cross-repo change-impact.
Live drift detection + drift gate	Two detectors: **contract drift** (an api_contract diverges from its consumers) and **pointer/hash drift** (a SoR artifact's external content changed since we indexed it). On detection the platform writes a **DriftRecord**, computes **change-impact via the traceability graph** (every downstream consumer/repo), and opens a decide_drift gate — architect for contract, tech_lead for pointer/hash. **Accept** re-indexes; **reject** flags for revert; nothing is silently written to a system of record. The signed Git push webhook is the live trigger (built on the Phase-2 reconcile).

Phase 4 — capstone slice (what this phase proves)

Onboard a **split FE/BE** project -> register-don't-recreate + the built-in code-intel running for real in a companion worktree -> a **gap report** with a resume phase -> the backend exports an api_contract, the frontend reads it read-through and records a cross-repo implements edge -> **resume mid-pipeline** triggers the next phase to its gate -> a **contract change opens a drift gate** whose change-impact **spans both repos**, decided by the architect. Verified end-to-end; the full platform suite is green across all phases.

3. The Local Execution & Testing Guide

This is a hands-on guide for a developer to run the platform locally and *see* every workflow work — in the browser, the API, the Temporal dashboard, and the object store. It is not about unit tests. Follow Part A to get the core running with zero external accounts; Part C adds the optional live connectors.

Before you start

Complete the prerequisites and credentials in **Section 4** first (Docker Desktop running, .env created, and — for live AI runs — an ANTHROPIC_API_KEY). On Windows without make, use the shim ./make.ps1 <target> in place of make <target>.

3.1 Part A — Bring up the platform (no external accounts needed)

Step 1 — Start the infrastructure

```
# from the repo root
docker compose up -d        # postgres (:5433), temporal (:7233 + UI :8080), minio (:9000 + console :900
docker compose ps              # all services should be 'running' / healthy
```

What you should see: open the **Temporal UI** at http://localhost:8080 (empty workflow list for now) and the **MinIO console** at http://localhost:9001 (login minioadmin / minioadmin; the sdlc-artifacts bucket already exists).

Step 2 — Migrate the database and seed the demo tenant

```
make migrate     # creates every table + its RLS policy (runs as the DB owner)
make seed        # seeds a demo tenant, project, profile, agent specs, connectors
```

What you should see: seed prints the created tenant/project ids and a connector count. (If make migrate reports an ownership error on an ALTER, run it as the postgres superuser — see Section 4.4.)

Step 3 — Run the application (API + worker + frontend)

```
make dev         # = docker compose --profile app up
#   backend API  -> http://localhost:8000
#   React UI     -> http://localhost:5173
#   worker       -> connects to Temporal (durable runs execute here)
```

What you should see: the React app loads at http://localhost:5173; the worker logs “started” and registers with Temporal. A quick API smoke check:

```
curl http://localhost:8000/api/projects/ -H "X-Tenant-ID: <demo-tenant-id>"
# -> JSON list including the seeded demo project
```

3.2 Part B — Drive the planning slice in the browser (Phase 0 end-to-end)

This is the core, no-external-accounts workflow. Open http://localhost:5173 and work left to right through the five tabs.

> **Cost note**
>
> Triggering a phase runs the **real Claude agent** via the ClaudeDriver, so it needs a valid ANTHROPIC_API_KEY in .env and incurs a small API cost per run. To keep cost minimal, trigger a single phase (Ideation) first; that is enough to see the whole gate -> publish -> traceability loop.

Tab / action	What to do	What you should see (the proof)
Tenant + Project bar	Create or select a tenant, then a project (the seeded demo is fine).	The five tabs unlock.
1. Project Profile	Fill description, tech stack, important context; Save.	A new profile version is saved; it becomes the agent's context next run.
2. Phase Console	Add a WorkItem (a feature stream), then click **Ideation**.	A Run appears in the Run Timeline and moves running -> awaiting_gate as you poll. In the Temporal UI (:8080) a new AgentRunWorkflow is visible and *waiting* at the gate.
3. Gate Approvals	Find the pending approval; approve it as a user holding the required role.	The run resumes to completed; the workflow finishes in the Temporal UI.
4. Artifacts	Look at the ideation artifact.	Its status flips draft -> published; its body is stored in MinIO (visible in the :9001 console under drafts/ and snapshots/).
5. Traceability	Open the lineage view.	A graph of artifacts and their derives/tests/... edges; once external SoRs are connected, nodes deep-link to the live Confluence/Jira items.

Repeat the trigger for **Requirements -> PRD -> Architecture** to walk the full planning chain. Each run reads the artifacts the previous phase published (proving the platform — not a local file — is the source of

truth), and each is independently gated.

3.3 Part C — Optional: exercise the live connectors

These steps need real (test) cloud accounts. Each is independent — do only the ones you have credentials for. Set the relevant variables in .env (Section 4) and make seed again so the demo tenant gets the connector, then restart make dev.

Atlassian — Confluence + Jira (Phase 1)

- Set ATLASSIAN_*, CONFLUENCE_SPACE_KEY, JIRA_PROJECT_KEY.
- **Greenfield:** approve the PRD gate -> the PRD **publishes as a Confluence page**; approve the backlog -> epics/stories **appear in Jira**. The Traceability tab now deep-links to those live items.
- **Brownfield:** POST a Word/Excel document to /api/source-documents/ -> it is extracted into reviewable internal artifacts (gated before adoption); existing Jira issues are registered without duplicates.

GitHub — the build path (Phase 2)

Set GITHUB_API_URL, GITHUB_TOKEN (a PAT with repo scope), GITHUB_REPO ("owner/name"), GITHUB_DEFAULT_BASE, GITHUB_WEBHOOK_SECRET. Then enroll a companion on your machine and run a build:

```
# 1) enroll this machine as a runner (prints a one-time token)
python -m runner.companion enroll --api-url http://localhost:8000 \
    --tenant <tenant-id> --project <project-id> --label my-laptop --repos owner/name

# 2) run an assigned build task in a managed worktree
python -m runner.companion run --task task.json \
    --repo-path ~/repos/your-clone --worktrees ~/sdlc
```

What you should see: a feature/<key> branch is created in a worktree, tests run locally, a **draft PR opens on GitHub**, code review runs, and the Temporal UI shows the BuildWorkflow parked at the **reviewer merge gate**. Approve it (Gate Approvals) and the PR **merges**. Push a manual commit to the branch and the signed webhook reconciles it (the artifact's sha pointer moves; no duplicate).

Figma + Xray (Phase 3)

- **Figma:** set FIGMA_TOKEN + FIGMA_FILE_KEY (a personal-access token + a test file). The design phase reads the *current* design tokens/components from that file (read-as-SoR) instead of inventing them.
- **Xray:** set XRAY_CLIENT_ID + XRAY_CLIENT_SECRET + XRAY_PROJECT_KEY. The test phase authors tests into Xray and imports results; a failing import opens a defect WorkItem linked to the test and traced to the code, which closes automatically on a clean re-run.

3.4 Part D — See the cross-cutting guarantees with your own eyes

Guarantee	How to observe it
Durability	In the Temporal UI (:8080), open a running workflow — it shows each activity, retries, and the gate wait. Restart the worker container mid-run; the workflow resumes where it left off.

Guarantee	How to observe it
Gate-then-publish	On the Artifacts tab, an artifact stays draft until you approve its gate — then it becomes published. There is no way to publish without an approval.
Tenant isolation (RLS)	Create a second tenant and call the API with its X-Tenant-ID: it sees none of the first tenant's projects, runs, artifacts, or defects.
Provenance / cost	The Run Timeline shows each run's engine, model, cost (USD), and Temporal workflow id; the Event ledger records who approved what and when.
Artifact bodies in S3	The MinIO console (:9001) shows artifact draft bodies and immutable post-approval snapshots — the registry holds only pointers + hashes.

> **"Everything works" checklist**
> (1) infra healthy; (2) migrate + seed succeed; (3) UI loads and the five tabs work; (4) a planning phase runs to completed through its gate and publishes an artifact; (5) traceability shows the lineage; (6) the Temporal UI shows the durable workflow; (7) MinIO shows the artifact body; (8) optional: a live PRD reaches Confluence, a backlog reaches Jira, a build reaches a GitHub PR, and a failing test opens a linked defect.

4. Setup, Credentials & Prerequisites

Everything a developer must install, configure, or obtain before running Section 3. Copy .env.example to .env and fill the values below.

4.1 Prerequisites to install

Tool	Why	Notes
Docker Desktop	Runs the whole stack (Postgres, Temporal, MinIO, API, worker, frontend).	Must be **running** before docker compose up. Windows: WSL2 backend recommended.
Python 3.12 + a repo-root .venv	Local tooling: fast pytest/ruff, the companion runner CLI.	python -m venv .venv then install backend/requirements-dev.txt. Tooling only — the app runs in Docker.
Git	Version control + the companion's managed worktrees.	Any recent version.
make (optional)	Convenience targets.	Windows without make: use ./make.ps1 <target>.

4.2 Environment variables (.env)

Required for the core local run

Variable	Purpose	Default / value
DJANGO_SECRET_KEY	Django signing key (dev).	any dev string
POSTGRES_* / DATABASE_URL	App database + the non-superuser RLS role.	preset in .env.example (app uses sdlc_app; host port 5433)

Variable	Purpose	Default / value
TEMPORAL_HOST / NAMESPACE / TASK_QUEUE	Durable workflow engine.	temporal:7233 / default / sdlc-planning
CORS_ALLOWED_ORIGINS	Lets the React app call the API.	http://localhost:5173
S3_* (MinIO)	Object store for artifact bodies + snapshots.	minioadmin / minioadmin, bucket sdlc-artifacts
ANTHROPIC_API_KEY	**Required to actually run agents** (the ClaudeDriver). Each run costs a little.	sk-ant-... from console.anthropic.com
DEFAULT_PLANNING_MODEL	Which Claude model planning runs use.	claude-opus-4-8 (use a cheaper model to reduce cost)

Optional — only for the live connector walkthroughs (Part C)

Variable group	Enables	Where to get it
ATLASSIAN_SITE_URL, ATLASSIAN_EMAIL, ATLASSIAN_API_TOKEN, ATLASSIAN_CLOUD_ID, CONFLUENCE_SPACE_KEY, JIRA_PROJECT_KEY	Phase 1: PRD -> Confluence, backlog -> Jira, traceability deep-links, brownfield onboarding.	API token: id.atlassian.com -> Security -> API tokens. Use a TEST space/project.
GITHUB_API_URL, GITHUB_TOKEN, GITHUB_REPO, GITHUB_DEFAULT_BASE, GITHUB_WEBHOOK_SECRET	Phase 2: companion build -> draft PR -> review -> merge gate -> merge; webhook reconcile.	A PAT with repo scope (github.com -> Settings -> Developer settings). Add these keys to .env (not in the template by default).
FIGMA_TOKEN, FIGMA_FILE_KEY	Phase 3: read the current design as the source of truth.	Figma -> Settings -> Personal access tokens; a test file's key from its URL.
XRAY_CLIENT_ID, XRAY_CLIENT_SECRET, XRAY_PROJECT_KEY	Phase 3: author tests + import results; failing test -> linked defect.	Xray Cloud -> Global Settings -> API Keys. Use a TEST project.

> **Security**
>
> Real credentials live only in .env, which is git-ignored — never commit them. Always use **test** spaces/projects/repos, not production ones. Without any connector creds the core local run (Parts A & B) still works fully on the internal source of truth.

4.3 Ports reference

Service	URL	Login
React UI (Experience plane)	http://localhost:5173	—
Backend API	http://localhost:8000/api	X-Tenant-ID header
Temporal Web UI	http://localhost:8080	—
MinIO console (object store)	http://localhost:9001	minioadmin / minioadmin

Service	URL	Login
Postgres (host-mapped)	localhost:5433	postgres / postgres (admin)

4.4 Manual setup notes & gotchas

- **Migrate as the DB owner.** The app connects as the non-superuser sdlc_app role (so RLS applies). Schema changes (ALTER) must run as the postgres superuser. If make migrate fails on an ALTER, run migrations with the superuser DSN: DATABASE_URL=postgres://postgres:postgres@localhost:5433/sdlc_platform.
- **Postgres is on host port 5433**, not 5432, to avoid clashing with a native Postgres. In-container services still use postgres:5432.
- **If the stack drops** (a “Socket is not connected” error on :5433 usually means Docker Desktop stopped), start Docker Desktop and run docker compose up -d again.
- **Windows without make:** use ./make.ps1 <target> (e.g. ./make.ps1 dev).
- **First frontend start** runs npm install inside the container — the very first make dev takes a few minutes while node modules download.
- **The companion runner** runs on the developer's own machine (not in Docker) via python -m runner.companion ...; it connects outbound to the platform API.

> **Quick start, condensed**
>
> cp .env.example .env (set ANTHROPIC_API_KEY) -> docker compose up -d -> make migrate -> make seed -> make dev -> open http://localhost:5173 and run a phase through its gate.

This document describes Phases 0–4 (complete and tested). Section 5 is an explicit test-through inventory of what works today. The next milestone is Phase 5 — Enterprise Hardening & Cloud Migration (RBAC + segregation of duties, secrets vault, prompt-injection hardening, immutable audit/compliance, the eval harness, and the cloud runner fleet).

5. End-to-End Testing — Two Full Journeys

This section is for an engineer running a real, straight-through test pass with the live credentials configured — not slice by slice. Each journey runs the whole lifecycle end to end: **Journey A** takes a brand-new (greenfield) project from an idea to a merged, tested change; **Journey B** onboards an existing (brownfield) project and resumes it from the middle. Complete **Section 3** (bring-up) once first, then follow the preflight below.

> **Conventions used in every command**
>
> All API calls carry -H "X-Tenant-ID: $TENANT" (the only exception is creating a tenant). Human gates are shown as a browser action on the **Gate Approvals** tab, with the equivalent POST /approvals/{id}/decide/ next to it. A gate decision must be made by a user who **holds the gate's role** on the project — the preflight grants the demo user every role so one person can approve the whole run. Phase-4 brownfield services have no dedicated UI yet, so Journey B uses a few manage.py shell calls (clearly marked) alongside the REST routes.

5.1 Preflight (live mode) — do this once

Put the live credentials in .env (see Section 4.2: Atlassian, GitHub, Figma, Xray, plus ANTHROPIC_API_KEY), then re-seed so the demo tenant picks up the connectors, and restart the app:

```
cp .env.example .env          # then fill the live values from Section 4.2
docker compose up -d          # infra: postgres:5433, temporal:7233/8080, minio:9001
make migrate && make seed     # seed attaches the connectors to the demo tenant
make dev                      # API :8000, React UI :5173, Temporal worker
```

Set a few shell variables and grant the seeded demo user every gate role (so a single approver can pass every gate in the run):

```
export API=http://localhost:8000/api
export TENANT=00000000-0000-0000-0000-0000000000d1   # the seeded demo tenant
export H="-H X-Tenant-ID:$TENANT -H Content-Type:application/json"
# the demo user (demo@demo.test) seeds with product_owner+architect; add the rest:
docker compose exec backend python manage.py shell -c "\
from apps.tenancy.models import Membership, User; from apps.projects.models import Project;\
u=User.objects.get(email='demo@demo.test');\
[Membership.objects.get_or_create(tenant_id=p.tenant_id,user=u,project=p,role=r) \
 for p in Project.objects.all() \
 for r in ['product_owner','architect','designer','tech_lead','reviewer','qa_lead','security_engineer','
```

> **Get the demo user id for gate decisions**
>
> curl -s $API/../ — the gate-decision body needs the approver's user_id. Fetch it once: docker compose exec backend python manage.py shell -c "from apps.tenancy.models import User; print(User.objects.get(email='demo@demo.test').id)" and export it as USER.

5.2 Journey A — Greenfield (new project, full lifecycle)

A new project from scratch, idea through a merged + tested change. Steps 1–3 create the project, its profile, and a work item; copy the returned ids into the shell variables.

```
# 1) create a greenfield project
curl -s $H -X POST $API/projects/ -d '{"name":"Acme Shop","track":"greenfield"}'
export PROJECT=<id-from-response>

# 2) set the project profile (becomes the agents' context)
curl -s $H -X PUT $API/projects/$PROJECT/profile/ -d '{"description":"A web shop","tech_stack":{"backend
```

```
# 3) create a feature WorkItem
curl -s $H -X POST $API/work-items/ -d '{"project":"'$PROJECT'","type":"story","feature_stream":"checkou
export WI=<id-from-response>
```

Now walk the lifecycle. Each phase is the *same* two moves — **trigger** the agent on the work item, then **approve** its gate — so the pattern is shown once and the per-phase agent keys, gate roles, and live proofs follow in the table:

```
# the repeating pattern (run once per phase, substituting AGENT below):
curl -s $H -X POST $API/work-items/$WI/trigger/ -d '{"agent_key":"01_ideation"}'
#   -> 202 with a run_id; the run moves running -> awaiting_gate
curl -s $H "$API/runs/?work_item=$WI"            # watch status; also Temporal UI :8080
curl -s $H "$API/approvals/?decision=pending&work_item=$WI"   # get the approval id
curl -s $H -X POST $API/approvals/<approval-id>/decide/ \
     -d '{"user_id":"'$USER'","decision":"approved"}'   # or the Gate Approvals tab
```

#	Phase — agent_key	Gate role	What you should see (the live proof)
1	Ideation — 01_ideation	product_owner	Run completes; an ideation artifact flips draft -> published (Artifacts tab / MinIO body).
2	Requirements — 02_requirements	product_owner	A requirements artifact published; Traceability shows it derives from ideation.
3	PRD — 03_prd	product_owner	**A Confluence page is created** on approval; the artifact's external_ref.url and the Traceability node deep-link to it.
4	Architecture — 04_architecture	architect	An architecture (+ ADR/data-model/api-spec) artifact set published.
5	UX Design — 05_ux_design	designer	**Reads your live Figma** (tokens/components); the **WCAG gate** blocks approval until every required criterion is satisfied (send them in the decide call's conditions, or tick them in the UI).
6	Task Breakdown — 06_task_breakdown	tech_lead	**Epics/stories appear in Jira** on approval; child WorkItems are created and Traceability links story -> requirement.
7	Implementation — 07_implementation	tech_lead	Runs on the **companion** (next code block) -> a **draft PR opens on GitHub**; the BuildWorkflow is visible in the Temporal UI.
8	Code Review — 08_code_review	reviewer	Review findings become child WorkItems; the 07 <-> 08 remediation loop re-runs until clean.
9	Testing — 09_testing (+ 09b_test_automation)	qa_lead	**Tests author to Xray + results import**; a failing test opens a linked, traced defect WorkItem that closes on a clean re-run.
10	Security — 10_security	security_engineer	A security review + **SBOM**; any Critical/High finding opens a defect and blocks the gate until resolved.

#	Phase — agent_key	Gate role	What you should see (the live proof)
11	Merge	reviewer	Approve the **reviewer merge gate** -> the GitHub PR **merges**; the WorkItem moves to done.
12	CI/CD + Deploy — 11_cicd, 12_deployment	devops	Pipeline config + a release/deployment artifact are produced (real cloud deploy is Phase 5).

The build step (phase 7), on your machine. After triggering 07_implementation, run the companion against your real clone — it works in a platform-managed worktree branch, builds/tests locally, and opens the PR:

```
# enroll this machine once (prints a one-time token)
python -m runner.companion enroll --api-url http://localhost:8000 \
    --tenant $TENANT --project $PROJECT --label my-laptop --repos owner/name

# fetch the assigned build task for the run, then execute it in a managed worktree
curl -s $H "$API/runner/builds/<run_id>/" > task.json
python -m runner.companion run --task task.json \
    --repo-path ~/repos/your-clone --worktrees ~/sdlc
```

> **Greenfield journey — done**
>
> You have taken a brand-new project idea -> requirements -> **PRD in Confluence** -> architecture -> **UX from live Figma + WCAG gate** -> **backlog in Jira** -> **a built, reviewed GitHub PR** -> **tests in Xray with a linked defect** -> a security review -> **merge** -> CI/CD + deploy artifacts. Every transition was human-gated and recorded in the Event ledger and Run timeline.

5.3 Journey B — Brownfield (existing project, resume mid-pipeline)

Onboard a real existing project, let the platform tell you where to resume, then carry on. The Phase-4 onboarding / API-contract / drift services have no dedicated UI yet, so this journey mixes the REST routes that exist with a few marked manage.py shell calls; the human gates are still approved in the browser. (The single self-contained proof of the whole flow is pytest tests/test_acceptance_phase4.py -v.)

```
# 1) create a brownfield project, then register its repositories (one or many)
curl -s $H -X POST $API/projects/ -d '{"name":"Legacy Shop","track":"brownfield"}'
export PROJECT=<id-from-response>
docker compose exec backend python manage.py shell -c "\
from apps.projects.repos import add_repository as A;\
print(A(tenant_id='$TENANT',project_id='$PROJECT',name='owner/web',role='frontend'));\
print(A(tenant_id='$TENANT',project_id='$PROJECT',name='owner/api',role='backend'))"
#   -> repo_topology reconciles to split/multi
```

```
# 2) code intelligence (00c): run the built-in analyzer on your clone, register it,
#    and index it into the Knowledge Index (platform stores intel + chunks, not source)
docker compose exec backend python manage.py shell -c "\
from integrations.codeintel.builtin import BuiltinCodeIntel;\
from mcp_platform_state import tools; from apps.knowledge.index import index_artifact;\
idx=BuiltinCodeIntel().index(workspace='/path/to/your/clone', repo='owner/api');\
print(idx.file_count, idx.commits, idx.hotspots);\
a=tools.register_artifact('$TENANT','$PROJECT',None,'context_extraction','codebase_intelligence','interr
index_artifact(tenant_id='$TENANT',project_id='$PROJECT',artifact_id=a['id'],text='backend exposes GET /
```

```
# 3) register what already exists (register-don't-recreate) + adopt static docs
curl -s $H -X POST $API/projects/$PROJECT/onboard-atlassian/   # existing Confluence/Jira
curl -s -H "X-Tenant-ID:$TENANT" -F file=@./old-spec.docx \
     $API/projects/$PROJECT/source-documents/                  # extract -> draft
curl -s $H -X POST $API/source-documents/<id>/adopt/ -d '{"user_id":"'$USER'"}'  # gate
```

```
# 4) gap report -> the resume phase; 5) cross-repo API-contract read-through
docker compose exec backend python manage.py shell -c "\
from integrations.onboarding import build_gap_report;\
from integrations.api_contract import export_api_contract, read_api_contract, record_implements;\
r=build_gap_report(tenant_id='$TENANT',project_id='$PROJECT');\
print('RESUME PHASE ->', r['resume_phase'], '| gaps:', r['blockers']);\
c=export_api_contract(tenant_id='$TENANT',project_id='$PROJECT',body='GET /users',source_commit='abc123'
print('contract', c['semver']);\
print('read-through', read_api_contract(tenant_id='$TENANT',project_id='$PROJECT')['external_ref']['semv
```

6) Resume mid-pipeline. Trigger the phase the gap report named (e.g. if it says architecture, trigger 04_architecture on a work item) and approve its gate in the browser — onboarding has handed straight to the same pipeline Journey A used.

```
curl -s $H -X POST $API/work-items/$WI/trigger/ -d '{"agent_key":"04_architecture"}'
#   then approve the architect gate (Gate Approvals tab / the decide call from 5.2)
```

```
# 7) contract change -> a drift gate with cross-repo change-impact -> architect decides
docker compose exec backend python manage.py shell -c "\
from integrations.api_contract import export_api_contract;\
from integrations.drift import detect_contract_drift, open_drift_records, decide_drift;\
b=export_api_contract(tenant_id='$TENANT',project_id='$PROJECT',body='GET /users (v2 breaking)',change='
d=detect_contract_drift(tenant_id='$TENANT',contract_artifact_id=b['artifact_id'],new_semver='2.0.0');\
print('impacted repos/artifacts ->', d['impacted']);\
print(decide_drift(tenant_id='$TENANT',drift_id=d['drift_id'],user_id='$USER',decision='accept'))"
#   (or trigger it live: push to the provider repo so the signed GitHub webhook fires)
```

Brownfield journey — done

You onboarded an existing multi-repo project, indexed its code, registered its existing Confluence/Jira + static docs without duplicates, got a **gap report with a resume phase**, exported a **versioned API contract read across repos**, **resumed mid-pipeline** into the normal gated flow, and watched a **contract change open a drift gate** whose impact spanned both repos. Re-running any register step adds nothing (register-don't-recreate).

5.4 If something stalls

Almost always it is the environment, not the platform. Quick checks:

- **Infra down:** a “Socket is not connected” on :5433 means Docker Desktop stopped — start it and docker compose up -d again.
- **Migrations:** if make migrate errors on an ALTER, run it with the postgres superuser DSN (Section 4.4).
- **No agent output / 500 on trigger:** ANTHROPIC_API_KEY is missing or the model id is wrong (Section 4.2).
- **Live SoR step does nothing:** the connector creds aren't set, or you didn't make seed after adding them — reseed and restart make dev.

- **Gate won't approve (403):** the user_id doesn't hold that gate's role on the project — re-run the role-grant snippet in 5.1.
- **Full automated proof anytime:** cd backend && pytest -q (**423 passed, 3 skipped**; the 3 skips are external-SaaS live round-trips).